$\mathcal{E}_i = -\frac{d\varphi}{dt}$; $d\varphi = B\,dS$, $dS = dy \cdot 2x$

$d\varphi = B\,dy \cdot 2x$, $a = \frac{dv}{dt}$

$\varphi = \frac{2B}{\sqrt{k}} \int \sqrt{y}\,dy = \frac{2B}{\sqrt{k}} \cdot \frac{y^{\frac{3}{2}}}{\frac{3}{2}} + C^{=0}$

$\int dv = \int a\,dt$;

$v = at + C = 0$

$v = \frac{dy}{dt}$; $dy = at\,dt$

$y = a\int t\,dt = \frac{at^2}{2}$

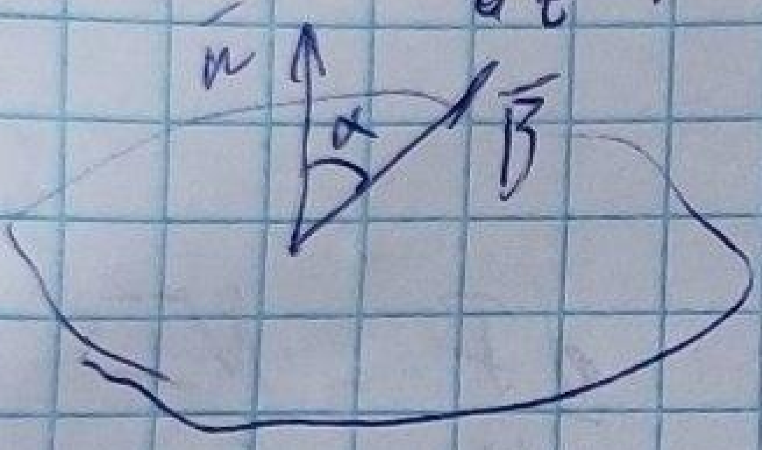
$\varphi(t) = \frac{2B}{\sqrt{k}} \cdot \frac{2(at^2)^{\frac{3}{2}}}{2^{\frac{3}{2}} \cdot 3} = \frac{\sqrt{2}\,B t^3 a^{\frac{3}{2}}}{\sqrt{k} \cdot 3}$

$$\int_0^{\Phi} d\Phi = \int_a^{a+b} \frac{A}{x}\,\frac{(cb + ac - cx)\,dx}{b}$$

$$\Phi = \frac{A}{x}\left(\int_a^{a+b} \frac{bc}{x}\,dx + \int_a^{a+b} \frac{ac\,dx}{x} - \int_b^{a+b} c\,dx\right) =$$

$$= \frac{A}{x}\left(bc \ln x\Big|_a^{a+b} + ac\ln x\Big|_a^{a+b} + cx\Big|_a^{a+b}\right)$$

$\mathcal{E}_i = -\frac{d\Phi}{dt}$, a, R, α, B; Q — ? (своб. конт. к.)

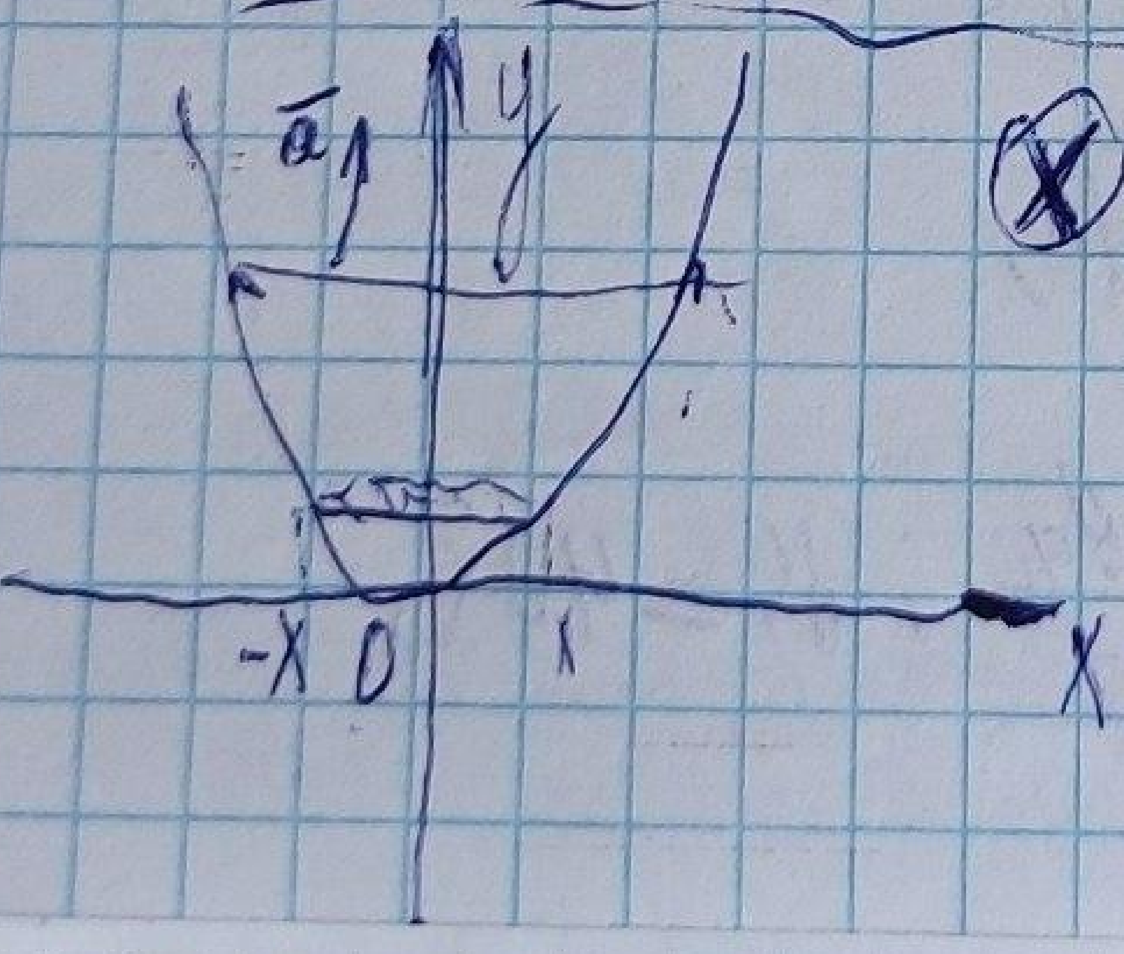


$$\begin{cases} \mathcal{E} = -\frac{\Delta\Phi}{\Delta t} \\ I = \frac{\mathcal{E}}{R} \\ I = \frac{d\Phi}{dt\,R} = \frac{dq}{dt} \end{cases}$$

$\Phi = Ba^2\cos\alpha$

$dq = I\,dt = \frac{\mathcal{E}}{R}\,dt = -\frac{d\Phi}{R}$

$$\int_0^{Q} dq = \int_{\Phi_0}^{0} \frac{1}{R}\,d\Phi = \frac{1}{R}\cdot\left(Ba^2\cos\alpha - 0\right) = \frac{Ba^2\cos\alpha}{R}$$



$\otimes\ \vec{B}$

$y = kx^2$; $y(0) = 0$; a, B

$v(0) = 0$, $\mathcal{E}(y)$ — ?